

Requivo AI

Autonomous ERP Operations Agent

Technical Specification & Acceptance Criteria Document

Document Version	v1.0
Status	Draft — Pending Review
Classification	Internal — Confidential
Date	June 2026
Prepared By	Lumenware Technologies Team

Document Control

Version History

Version	Date	Author	Description
1.0	June 2026	Engineering Team	Initial draft
—	—	—	Pending stakeholder review

Reviewers & Approvers

Role	Name	Signature	Date
Engineering Lead	Goodness Dare	_____	
Product Owner	Goodness Dare	_____	
Senior Fullstack Developer	Excel Shogbola	_____	
Fullstack Developer (Intern)	Jasayo Rabi	_____	
Security Lead	TBD	_____	
QA Lead	Jasayo Rabi	_____	

Related Documents

- Requivo Architecture Diagram
- API Reference Guide
- Security & Compliance Policy
- Test Plan & Test Cases
- Deployment Runbook

Introduction

1.1 Purpose

This document defines the technical specifications and acceptance criteria for Requivo AI — an autonomous ERP operations agent that converts natural language business requests into structured, executable multi-step workflows across procurement, inventory, sales, finance, and HR systems. It serves as the authoritative reference for engineering, QA, product, and stakeholder teams during development, testing, and sign-off.

1.2 Scope

This document covers:

- System architecture and component-level technical specifications
- Interface contracts for all business tool integrations
- Data model requirements and persistence strategy
- Security, compliance, and RBAC requirements
- Human-in-the-Loop (HITL) workflow design
- Observability, logging, and monitoring standards
- Functional and non-functional acceptance criteria per module
- Definition of Done and sign-off checklist

1.3 Audience

Audience	Usage
Engineering Team	Build to spec; understand component contracts
QA / Test Engineers	Derive test cases from acceptance criteria
Product Owner	Validate scope and business requirements
Security Team	Review compliance and access control requirements
Stakeholders	Baseline for milestone sign-off

1.4 Definitions & Abbreviations

Term	Definition
ERP	Enterprise Resource Planning
LLM	Large Language Model
HITL	Human-in-the-Loop — manual approval gate in automated workflow
RBAC	Role-Based Access Control
PO	Purchase Order
Workflow	A multi-step automated task orchestrated by Requivo
Tool	A domain-specific ERP API adapter exposed to the agent
AC	Acceptance Criteria

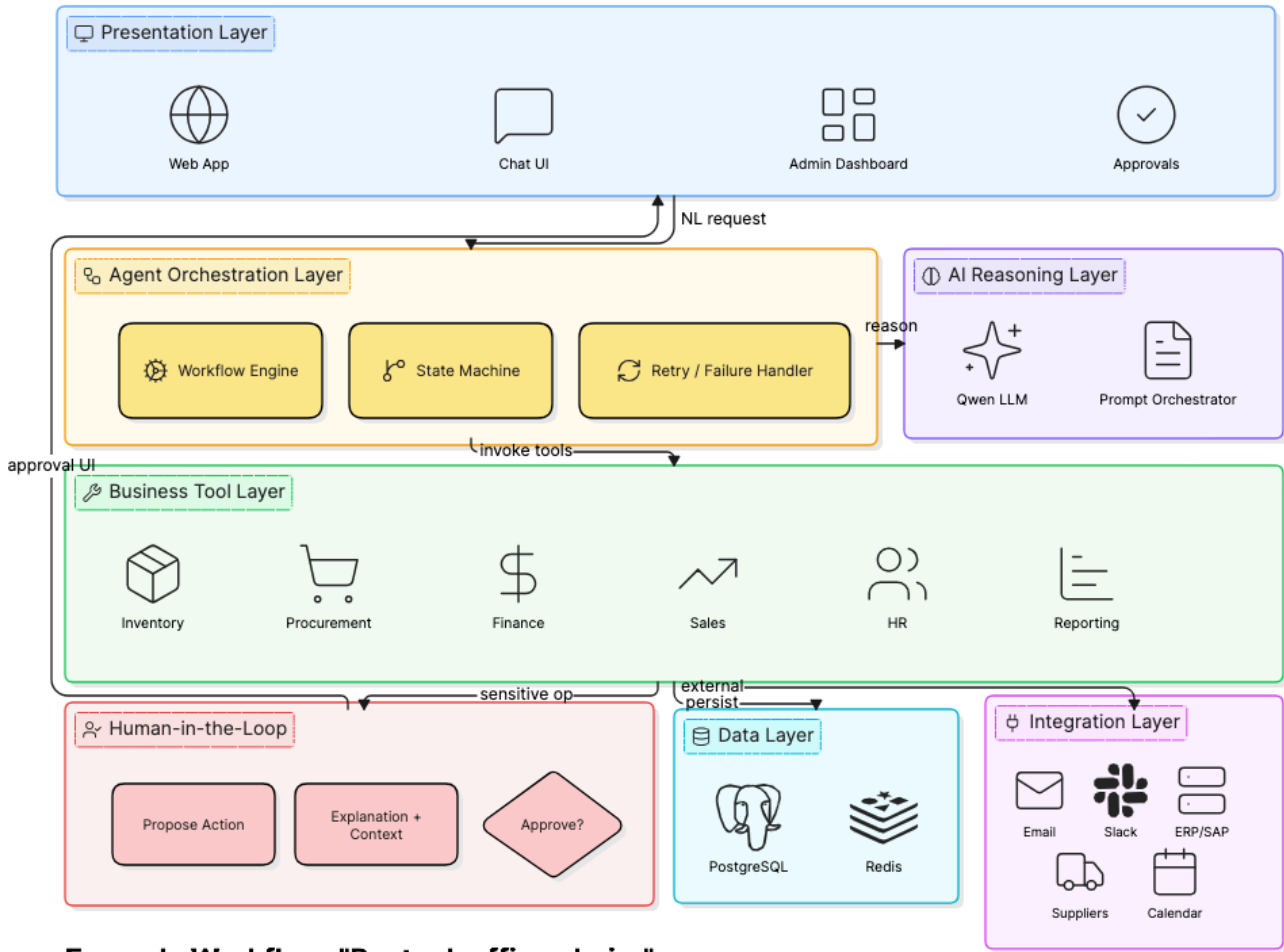
Term	Definition
DL Queue	Dead-Letter Queue — storage for unrecoverably failed workflows
Redis	In-memory data store used for state caching

System Architecture

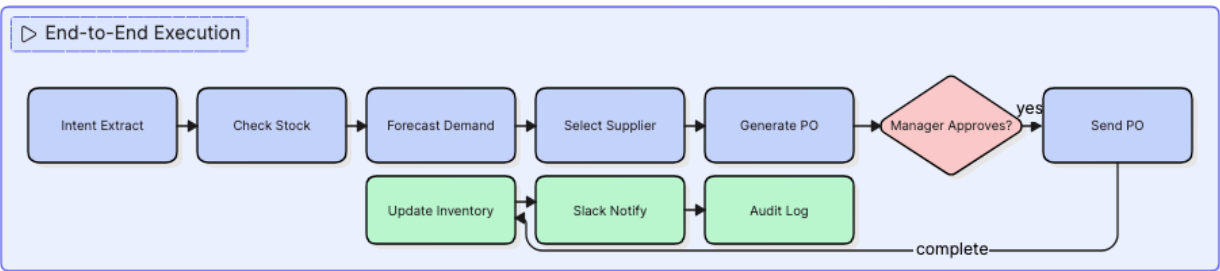
2.1 Architecture Overview

Requivo follows a layered, service-oriented architecture with clear separation of concerns across six layers. Each layer has a defined responsibility and communicates through typed interfaces.

ERPilot Architecture & Workflow



Example Workflow: "Restock office chairs"



Layer	Technology Stack	Primary Responsibility
Presentation	React.js, TypeScript	User interface, chat input, approvals dashboard, audit display
Agent Orchestration	ASP.NET Core ASP.NET MVC	Workflow state machine, task decomposition, retry logic
AI Reasoning	Qwen LLM, Prompt Orchestration	Intent extraction, planning, tool selection, decision-making
Business Tool Layer	Domain-specific APIs	ERP abstraction: Inventory, Procurement, Finance, Sales, HR, Reporting
HITL Layer	Approval Engine	Pause, context surfacing, human decision capture, audit write
Data Layer	PostgreSQL, Redis	Entity persistence, workflow state, agent memory, audit logs

2.2 Component Specifications

2.2.1 Presentation Layer

Attribute	Specification
Framework	React with TypeScript
Interface type	Chat-based natural language input + Admin dashboard
State management	React context or Zustand for UI state
Real-time updates	WebSocket or Server-Sent Events (SSE) for workflow progress
Auth integration	OAuth 2.0 / SSO via presentation layer middleware
Accessibility	WCAG 2.1 AA compliance

2.2.2 Agent Orchestration Layer

Attribute	Specification
Runtime	.NET 10
Workflow engine	Custom state machine with defined state transitions
Task execution	Sequential and parallel tool invocation supported
State persistence	Redis for active workflow state; PostgreSQL for completed
Retry policy	Exponential backoff: 3 attempts, delays 1s / 5s / 30s
Timeout	Per-task: 30s; full workflow: configurable (default 10 min)
Concurrency	Up to 50 concurrent workflows per deployment instance

2.2.3 AI Reasoning Layer

Attribute	Specification
Core model	Qwen LLM (hosted inference endpoint)
Context window	Minimum 32k tokens; 128k preferred for complex workflows
Prompting strategy	System prompt + tool schemas + memory injection per request
Fallback	Escalate to human if LLM confidence below threshold (< 0.7)
Max retries on LLM error	2 retries with prompt reformulation
Response format	Structured JSON for tool calls; natural language for explanations
Latency target	P95 < 5 seconds per LLM call

2.2.4 Data Layer

Component	Purpose	Technology
Primary Store	Business entities, audit logs, approval history	PostgreSQL 18+
Cache / State Store	Active workflow state, session data, agent context snapshots	Redis 7+
Encryption at rest	AES-256 for sensitive fields (PII, financial data)	Database-level + field-level
Encryption in transit	All service-to-service communication	TLS 1.3
Backup frequency	Full daily backup; transaction log shipping every 15 min	Automated via infrastructure
Retention	Audit logs: 7 years; Workflow data: 3 years	Tiered storage policy

2.3 Workflow State Machine

All workflows are modelled as a deterministic state machine with the following states and valid transitions:

State	Description	Valid Next States
PENDING	Workflow received, not yet started	PLANNING, FAILED
PLANNING	LLM decomposing request into tasks	IN_PROGRESS, WAITING_APPROVAL, FAILED
WAITING_APPROVAL	Paused — awaiting human decision	IN_PROGRESS, FAILED
IN_PROGRESS	Tools executing step-by-step	WAITING_APPROVAL, COMPLETED, FAILED
COMPLETED	All steps finished successfully	(terminal)
FAILED	Unrecoverable error or rejection	(terminal)

2.4 Human-in-the-Loop (HITL) Design

The HITL layer pauses execution at defined trigger points and surfaces a rich decision context to the approver.

Trigger Conditions

Trigger	Threshold / Condition	Escalation Level
Financial transaction	> \$1,000 (configurable per org)	Line manager
Vendor creation / modification	Any new vendor record	Procurement lead
Purchase order execution	Any PO issuance	Finance approver
Employee lifecycle actions	Onboarding, offboarding, payroll changes	HR lead
Bulk data modification	> 100 records in single action	System admin
Low AI confidence	LLM confidence score < 0.7	Assigned operator

Interface Specifications

3.1 Business Tool Interface Contract

Every domain tool exposed to the agent must implement the following TypeScript interface:

```
using System.Collections.Generic;
using System.Threading.Tasks;

namespace AgentFramework.Tools
{
    public interface ITool
    {
        string Name { get; }
        string Description { get; }
        JsonSchema InputSchema { get; }

        Task<ToolResult> ExecuteAsync(
            object? input,
            WorkflowContext context);
    }

    public class ToolResult
    {
        public bool Success { get; set; }

        public object? Data { get; set; }

        public string? Error { get; set; }

        public ToolMetadata? Metadata { get; set; }
    }

    public class ToolMetadata
    {
        public long Latency { get; set; }

        public string Source { get; set; } = string.Empty;
    }
}
```

```
public class WorkflowContext
{
    public string WorkflowId { get; set; } = string.Empty;

    public string UserId { get; set; } = string.Empty;

    public List<string> Permissions { get; set; } = new();

    public int StepIndex { get; set; }

    public List<AuditEntry> AuditTrail { get; set; } = new();
}

// Placeholder types
public class JsonSchema
{
}

public class AuditEntry
{
}
}
```

3.2 Domain Tool Specifications

Tool	Key Operations	Critical Input Fields	Approval Required
Inventory Tool	Check stock, update levels, forecast demand, set thresholds	SKU, quantity, location, forecast_period	Bulk updates > 100 units
Procurement Tool	Supplier selection, PO creation, PO submission	supplier_id, items[], amount, delivery_date	All PO issuances
Finance Tool	Invoice creation, expense logging, payment initiation	amount, account_code, vendor_id, due_date	Payments > \$1,000
Sales Tool	Quote generation, order creation, pricing update	customer_id, items[], discount_pct	Discount > 20%
HR Tool	Onboarding, leave management, payroll updates, contract management, appraisals, interview	employee_id, action_type, effective_date	All lifecycle events

Tool	Key Operations	Critical Input Fields	Approval Required
Reporting Tool	KPI dashboards, analytics queries, export	report_type, date_range, filters	Never

3.3 Integration Layer Specifications

Integration	Protocol	Auth Method	SLA / Timeout
Email (notifications)	SMTP / SendGrid API	API Key	Async; best-effort
Slack (alerts)	Slack Web API	Bot OAuth Token	Async; < 3s delivery
Legacy ERP systems	REST / SOAP adapters	API Key / Basic Auth	< 10s per call
QuickBooks / SAP-like	REST API	OAuth 2.0	< 5s per call
Supplier APIs	REST / EDI	API Key (per supplier)	< 10s per call
Calendar systems	Google Calendar / Exchange API	OAuth 2.0	< 3s per call

Security & Compliance Requirements

4.1 Access Control

Requirement	Specification
Authentication	Multi-factor authentication (MFA) mandatory for all users
Authorization	RBAC with principle of least privilege enforced at action level
Session management	JWT with 1-hour expiry; refresh token rotation on use
API authentication	Bearer token required on all internal service calls
Service accounts	Dedicated service accounts for each tool; no shared credentials
Audit logging	All actions (read + write) logged with user, timestamp, and outcome

4.2 RBAC Permission Matrix

Role	Inventor y	Procurem ent	Finance	Sales	HR	Admin
System Admin	Full	Full	Full	Full	Full	Full
Finance Manager	Read	Approve	Full	Read	Read	None
Procurement Lead	Read/Wri te	Full	Read	Read	None	None
HR Manager	None	None	Read	None	Full	None
Sales Rep	Read	None	None	Read/Wri te	None	None
Auditor (read-only)	Read	Read	Read	Read	Read	None
AI Agent (system)	Per-work flow	Per-workfl ow	Per-work flow	Per-work flow	Per-work flow	None

4.3 Data Protection

Requirement	Specification
Encryption at rest	AES-256 for all database fields containing PII or financial data
Encryption in transit	TLS 1.3 minimum for all inter-service and external communication
PII handling	PII fields masked in logs; full access requires elevated permission
Data residency	All data stored within customer-defined geographic region
Right to erasure	API-level support for GDPR Article 17 deletion requests
Vulnerability scanning	Automated SAST/DAST on every CI build; critical issues block release
Penetration testing	Annual third-party pen test; critical findings resolved within 30 days

Performance & Reliability Requirements

Metric	Target	Measurement Method
System availability	99.9% uptime (< 8.7 hrs downtime/year)	Uptime monitoring
Workflow end-to-end latency (simple)	< 30 seconds P95	Workflow trace timestamps
Workflow end-to-end latency (complex)	< 5 minutes P95 (excluding approval wait)	Workflow trace timestamps
LLM inference latency	< 5 seconds P95 per call	LLM span in traces
Tool execution latency	< 10 seconds P95 per tool call	Tool span in traces
Approval notification delivery	< 60 seconds from HITL trigger	Notification timestamp delta
Concurrent workflow capacity	>= 50 simultaneous workflows per instance	Load test
Throughput	>= 500 workflow starts per hour	Load test
Database query latency	< 100ms P99 for indexed queries	DB slow query log
Error rate	< 1% of workflows ending in FAILED state (excl. rejections)	Monitoring dashboard

Acceptance Criteria

Acceptance criteria are organised by module. Each criterion is tagged with a priority:

- **MUST** — Mandatory for production release
- **SHOULD** — Strongly recommended; may be deferred with documented risk
- **MAY** — Optional enhancement

6.1 Workflow Orchestration Engine

AC-ID	Criterion	Test / Verification Method	Priority
AC-W01	The system accepts natural language input and successfully extracts workflow intent for all 6 supported ERP domains	Submit 20 representative NL requests; verify correct domain classification >= 95% accuracy	MUST
AC-W02	Workflows execute all defined state transitions without undefined state jumps	Unit + integration tests covering all valid and invalid transitions in the state machine	MUST
AC-W03	Workflow state is persisted to Redis after each step; system can resume from last checkpoint after restart	Kill process mid-workflow; restart; verify workflow continues from last completed step	MUST
AC-W04	Failed workflows are moved to the dead-letter queue with full execution history attached	Inject tool failures; verify DL queue entry contains full trace	MUST
AC-W05	The system supports >= 50 concurrent workflows without performance degradation	Load test with 50 concurrent workflows; verify all complete within SLA	MUST
AC-W06	Partial workflow rollback is executed for multi-step workflows that fail mid-execution	Trigger mid-flow failure after data mutation; verify compensating actions run	SHOULD

6.2 AI Reasoning Layer

AC-ID	Criterion	Test / Verification Method	Priority
AC-AI01	The LLM correctly decomposes at least 90% of well-formed business requests into valid task sequences	Evaluate 50 curated business requests against expected task sequences (human-graded)	MUST
AC-AI02	The agent asks clarifying questions when the request is ambiguous rather than proceeding with assumptions	Submit 10 intentionally ambiguous requests; verify clarification prompt is returned for >= 8	MUST
AC-AI03	LLM inference latency is < 5 seconds P95 under normal load	Run 100 inference calls; measure latency percentiles via tracing	MUST

AC-ID	Criterion	Test / Verification Method	Priority
AC-AI04	The agent provides a human-readable explanation for every proposed action before execution	Review explanation field in all workflow traces; verify non-empty for 100% of actions	MUST
AC-AI05	When LLM confidence is below 0.7, the workflow escalates to HITL rather than proceeding autonomously	Inject low-confidence scenarios; verify HITL trigger is raised	MUST
AC-AI06	Agent memory is correctly injected into subsequent steps for multi-turn context continuity	Run multi-step workflow; verify step N+1 references entities resolved in step N	MUST

6.3 Business Tool Layer

AC-ID	Criterion	Test / Verification Method	Priority
AC-T01	Every tool validates its input schema and returns a structured error (not an exception) for invalid inputs	Submit invalid inputs to each of the 6 tools; verify structured ToolResult error is returned	MUST
AC-T02	Tool execution latency is < 10 seconds P95 for all 6 tools under normal load	Benchmark each tool with 50 requests; verify P95 latency	MUST
AC-T03	Every tool write operation creates an audit log entry with: workflow ID, user ID, timestamp, input, output, and outcome	Execute one write per tool; inspect audit log for required fields on all 6 entries	MUST
AC-T04	The Inventory Tool correctly flags stock below configured threshold and includes forecast data in response	Set threshold; deplete stock below threshold; verify flag and forecast fields in response	MUST
AC-T05	The Procurement Tool generates a correctly structured PO document and routes it to the HITL approval queue	Initiate PO workflow; verify PO document structure and HITL queue entry	MUST
AC-T06	The Finance Tool enforces the payment approval threshold; transactions above threshold are blocked pending approval	Submit payment above threshold; verify transaction is suspended and approval notification sent	MUST
AC-T07	The HR Tool prevents employee lifecycle actions without a valid approval record	Attempt onboarding without approval; verify action is rejected with error	MUST
AC-T08	RBAC enforcement blocks tool access for roles without the required permission	Attempt tool invocations with underprivileged role tokens; verify 403 responses	MUST

6.4 Human-in-the-Loop (HITL)

AC-ID	Criterion	Test / Verification Method	Priority
AC-H01	All 5 defined HITL trigger conditions pause workflow execution and raise an approval request	Trigger each condition; verify workflow transitions to WAITING_APPROVAL	MUST
AC-H02	Approver receives notification (email and/or Slack) within 60 seconds of HITL trigger	Measure notification delivery time across 10 trigger events	MUST
AC-H03	Approval notification includes: workflow summary, proposed action, business context, and approve/reject controls	Inspect notification content for required fields on 5 sample approvals	MUST
AC-H04	Approved workflows resume execution correctly from the suspended step	Approve 5 paused workflows; verify each resumes and completes successfully	MUST
AC-H05	Rejected workflows transition to FAILED state with rejection reason recorded in audit log	Reject 5 paused workflows; verify FAILED state and audit entry with reason	MUST
AC-H06	Approval decisions are recorded with approver identity, timestamp, and rationale	Inspect audit log entries for approved/rejected workflows for all required fields	MUST
AC-H07	HITL dashboard displays all pending approvals with age, priority, and context	UI review against design spec; verify all required fields are visible	SHOULD

6.5 Security & Access Control

AC-ID	Criterion	Test / Verification Method	Priority
AC-S01	All API endpoints return 401 for unauthenticated requests	Send requests without valid token to all endpoints; verify 401 responses	MUST
AC-S02	RBAC enforcement is validated at the action level, not only at the API gateway	Bypass gateway; send authenticated requests for out-of-role actions; verify rejection	MUST
AC-S03	PII fields are masked in all application log outputs	Execute workflow with PII data; inspect logs; verify masking applied	MUST
AC-S04	All inter-service communication uses TLS 1.3 or higher	Network traffic inspection via automated TLS audit tool	MUST
AC-S05	SAST and DAST scans pass with zero critical or high findings before release	CI pipeline gate: SAST (Semgrep/Snyk) + DAST (OWASP ZAP) report inspection	MUST
AC-S06	Service account credentials are rotated and never hardcoded in source code	Static analysis + secrets scanning (e.g., Trufflesecurity); verify no hardcoded creds	MUST

6.6 Observability & Monitoring

AC-ID	Criterion	Test / Verification Method	Priority
AC-O01	Every workflow produces a structured execution trace: intent, tasks, tool calls, decisions, and outcome	Run 10 diverse workflows; inspect trace completeness in observability platform	MUST
AC-O02	Monitoring dashboards are available for: workflow success/failure rate, tool latency, approval queue depth, and system health	Access monitoring platform; verify all 4 dashboard types are populated with live data	MUST
AC-O03	Alerts fire within 5 minutes when workflow failure rate exceeds 5% over any 10-minute window	Inject artificial failures to trigger threshold; verify alert delivery time	MUST
AC-O04	LLM reasoning snapshots are stored and queryable for every workflow step	Query reasoning store for a completed workflow; verify all steps have snapshots	SHOULD
AC-O05	Tool execution latency is tracked per-tool and surfaced in the monitoring dashboard	Inspect dashboard; verify per-tool latency metrics are displayed and updated in real time	MUST

6.7 Integration Layer

AC-ID	Criterion	Test / Verification Method	Priority
AC-I01	Email notifications are delivered for all HITL events and workflow completions	Trigger 10 notification events; verify email delivery in test mailbox	MUST
AC-I02	Slack alerts are posted to the configured channel for approval requests and critical failures	Trigger alert events; verify Slack channel receives correctly formatted messages	MUST
AC-I03	Legacy ERP integration adapter correctly maps tool output to ERP API format with no data loss	Run data mapping test suite; compare input vs. ERP-received payload field by field	MUST
AC-I04	Integration failures do not crash the orchestration layer; failures are caught, logged, and escalated	Simulate integration downtime; verify graceful error handling and notification	MUST

Non-Functional Acceptance Criteria

7.1 Reliability

AC-ID	Criterion	Target
AC-NF01	System availability measured over 30-day rolling window	>= 99.9% uptime
AC-NF02	Recovery time objective (RTO) after infrastructure failure	< 15 minutes
AC-NF03	Recovery point objective (RPO) — maximum data loss window	< 15 minutes
AC-NF04	Failed workflow retry succeeds without manual intervention for transient errors	>= 90% of transient failures auto-recover

7.2 Scalability

AC-ID	Criterion	Target
AC-NF05	Horizontal scaling: add instance to handle increased load	New instance active within 3 minutes; no workflow disruption
AC-NF06	Database performance under 10x baseline load	P99 query latency < 500ms
AC-NF07	Redis memory usage under 1,000 concurrent workflow states	< 4 GB Redis memory consumption

7.3 Maintainability

AC-ID	Criterion	Target
AC-NF08	Code test coverage for orchestration and tool layers	>= 80% unit + integration coverage
AC-NF09	New tool can be added without modifying core orchestration code	Verified via dependency inversion audit
AC-NF10	All configuration externalised; no hardcoded environment-specific values	Static analysis + configuration review

End-to-End Test Scenarios

The following scenarios validate the complete system as integrated. Each must pass before production release.

Scenario	Input	Expected Outcome	Result
Restock procurement	"We need to restock office chairs."	10-step workflow completes: stock check → forecast → supplier select → PO generated → approval → PO sent → inventory updated → Slack notification → audit log created	[]
Finance payment	"Pay invoice INV-2041 for \$4,500 to Acme Corp"	Finance tool creates payment record; HITL triggered (> \$1,000); manager notified; on approval payment executes; audit log records approver identity	[]
HR onboarding	"Onboard Jane Smith as a Sales Engineer starting July 1"	HR tool creates employee record; system calendar updated; HITL triggered; on approval IT access provisioned; payroll entry created	[]
Ambiguous request	"Sort out the supply issue"	Agent returns clarifying question: asks for domain, specific item, and urgency	[]
Rejected approval	"Create a new vendor: Unknown Supplies Ltd"	Vendor creation paused; approver rejects; workflow transitions to FAILED; rejection reason logged in audit trail	[]
Partial failure recovery	Multi-step inventory update — step 3 fails due to tool timeout	Steps 1 & 2 outputs preserved; retry triggered for step 3; compensating action runs if retry exhausted	[]
RBAC enforcement	Sales Rep attempts to initiate a payment	Request blocked at tool layer with 403; audit log records unauthorized attempt	[]
Reporting query	"Show me procurement spend by supplier for last quarter"	Reporting tool returns structured KPI data; dashboard renders correctly; no HITL required	[]

Definition of Done

A feature or module is considered done when ALL of the following conditions are met:

#	Condition	Owner	Done?
1	All MUST-priority acceptance criteria pass	QA Lead	[]
2	Unit and integration test coverage >= 80%	Engineering Lead	[]
3	SAST + DAST scans clear with zero critical/high findings	Security Lead	[]
4	Load test executed at 2x expected peak throughput without breach of SLAs	Engineering Lead	[]
5	Audit logging verified for all write operations across all 6 tools	QA Lead	[]
6	RBAC matrix tested for all 7 defined roles	Security Lead	[]
7	All 8 end-to-end scenarios pass in staging environment	QA Lead	[]
8	Monitoring dashboards are live and alerts are configured	Platform Lead	[]
9	HITL approval flow tested end-to-end for all 5 trigger types	Product Owner	[]
10	User documentation and API reference updated	Technical Writer	[]
11	Production deployment runbook reviewed and approved	Engineering Lead	[]
12	Final sign-off obtained from Product Owner and Security Lead	Product Owner	[]

Sign-off

Role	Name	Signature	Date
Engineering Lead			
Product Owner			
QA Lead			
Security Lead			
Finance Stakeholder			